Problem Brief

Predicting Market Volatility Risk Using Liquidity Indicators and Machine Learning

Yang Liu

Objective

This project examines whether daily liquidity proxies from SPY and QQQ, together with the Cboe Volatility Index (VIX), help identify unusually large next-day movements in the US equity market. The objective is to compare an interpretable regularized logistic generalized linear model (GLM) with a random forest and translate the resulting risk scores into a practical monitoring rule.

End user

The intended user is a portfolio risk analyst or risk-monitoring team responsible for reviewing equity-market exposure. The output is an early-warning indicator for further investigation, not an automatic instruction to trade or hedge.

Decision context

After each trading day, the analyst uses information available up to that close to flag potential high-volatility risk on the following trading day. A positive event is defined in advance as an absolute next-day SPY close-to-close return exceeding 1.5%. Analysts can use the warning to prioritize a portfolio review, check concentration, or consider whether additional monitoring is justified.

Cost / loss framing

The decision exercise assigns an illustrative cost of five units to a missed high-risk day (false negative) and one unit to an unnecessary warning (false positive). This reflects a preference for avoiding missed stress events, but the units are not measured dollar losses or estimated trading costs. The operating threshold is selected using training-period out-of-fold predictions only; the 2025 test year is held out for evaluation.

Data and analytic scope

The study uses daily SPY and QQQ open/high/low/close/volume observations alongside VIX observations for 2018–2025. Liquidity is proxied by Amihud illiquidity and a Corwin–Schultz high-low spread estimate, not observed bid–ask quotes. The workflow includes K-means stress regimes, chronological cross-validation, an out-of-time 2025 test and a feature-ablation sensitivity check.

Expected use and limitation

The main value is a transparent risk-screening workflow rather than a fully reliable alert system. The selected baseline GLM obtained a 2025 ROC-AUC of approximately 0.804 but detected only 9 of 33 observed high-risk days at the selected threshold. Removing liquidity features did not reduce average validation ROC-AUC, so their incremental predictive contribution is uncertain. The rule should support, not replace, human judgment.